Overhead Absorption Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Overhead Absorption Efficiency (OAE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on OAE achieves an annualized gross (net) Sharpe ratio of 0.40 (0.39), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 54 (51) bps/month with a t-statistic of 4.62 (4.45), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage) is 22 bps/month with a t-statistic of 2.41.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding risk premia, empirical tests often struggle to identify firm characteristics that reliably capture heterogeneous exposure to aggregate consumption dynamics. We address this gap by examining how firms' overhead absorption efficiency—the ability to cover fixed costs through operational activity—relates to their sensitivity to consumption-based risk factors. Overhead absorption efficiency captures fundamental aspects of operating leverage that determine how firms transmit aggregate shocks to cash flows, making it a natural candidate for identifying cross-sectional variation in consumption risk exposure.

Firms with low overhead absorption efficiency face higher exposure to consumption risk through multiple channels that affect the intertemporal marginal rate of substitution. First, inefficient overhead absorption amplifies the transmission of aggregate consumption shocks to firm cash flows, as documented in models of operating leverage and business cycle risk Carlson et al. (2004). When consumption growth slows, firms with high fixed cost burdens relative to their ability to absorb these costs experience disproportionate deterioration in profitability, creating systematic covariation with marginal utility Zhang (2005). This mechanism becomes particularly pronounced during economic downturns when the marginal utility of consumption is high, generating state-dependent risk premia consistent with habit formation models Campbell and Cochrane (1999).

Second, overhead absorption efficiency determines firms' resilience to disaster risk and rare consumption disasters. Following Barro (2006) and Gabaix (2012), firms with poor overhead absorption face elevated bankruptcy risk during consumption disasters, as their inflexible cost structures prevent rapid adjustment to severe negative shocks. The option-like nature of equity claims implies that this disaster sensitivity translates directly into required returns, with investors demanding higher compensation for bearing tail risk Wachter and Seo (2019). The consumption-based interpretation suggests that overhead absorption efficiency proxies for firms' loading on long-run consumption risk factors that drive persistent variation in marginal utility.

Third, the overhead absorption mechanism interacts with time-varying risk aversion and precautionary savings motives that characterize the intertemporal marginal rate of substitution. Firms with inefficient overhead absorption exhibit higher sensitivity to uncertainty shocks that affect households' consumption-savings decisions Bansal and Yaron (2004). During periods of elevated macro uncertainty, these firms experience larger increases in discount rates as investors require additional compensation for bearing systematic risk that covaries with consumption volatility. The resulting return predictability reflects rational pricing of consumption risk exposure rather than behavioral biases or market frictions, consistent with the predictions of recursive utility models Epstein and Zin (1989).

We find strong evidence that overhead absorption efficiency predicts cross-sectional variation in expected returns, with economically and statistically significant risk premia that survive extensive robustness tests. The value-weighted long-short strategy that buys stocks with high overhead absorption efficiency and sells stocks with low efficiency generates average monthly returns of 41 basis points with a t-statistic of 3.17, corresponding to an annualized Sharpe ratio of 0.40. The strategy's alpha remains highly significant after controlling for conventional risk factors, ranging from 44 to 65 basis points per month across standard factor models including the Fama-French six-factor model, with t-statistics exceeding 3.38 in all specifications.

The overhead absorption efficiency premium exhibits remarkable stability across different portfolio construction methodologies and remains robust to transaction costs. Net of trading costs, the strategy achieves monthly returns of 39 basis points with a t-statistic of 3.06, while the generalized net alpha relative to the six-factor model equals 51 basis points with a t-statistic of 4.45. The premium persists within size quintiles, generating significant abnormal returns even among the largest stocks where market frictions are minimal—the strategy earns 37 basis points per month among firms above the 80th NYSE size percentile with a t-statistic of 2.52.

Controlling for related anomalies and conducting horse-race tests against the entire factor zoo confirms that overhead absorption efficiency contains unique information about expected returns. When we simultaneously control for the six most closely related predictors and the Fama-French six factors, the strategy maintains an alpha of 22 basis points per month with a t-statistic of 2.41. The overhead absorption efficiency signal ranks in the top quintile of anomaly performance, with its net Sharpe ratio exceeding 95% of documented return predictors, demonstrating that it captures a distinct dimension of consumption risk not spanned by existing factors.

Our study contributes to the consumption-based asset pricing literature by identifying overhead absorption efficiency as a powerful proxy for firms' exposure to systematic consumption risk, extending the theoretical insights of Carlson et al. (2004) and Zhang (2005) on operating leverage and expected returns. Unlike previous studies that focus on accounting-based measures of operating leverage, we demonstrate that overhead absorption efficiency—which captures the dynamic interaction between fixed costs and operational capacity—provides superior identification of consumption risk exposure. Our findings complement Novy-Marx (2011) on operating leverage and profitability by showing that the efficiency of overhead absorption, rather than its level, drives cross-sectional return variation through its connection to marginal utility dynamics.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023b) to establish the robustness and economic significance of the overhead absorption efficiency premium. This rigorous approach, which accounts for multiple testing concerns and transaction costs while controlling for the full factor zoo, provides compelling evidence that overhead absorption efficiency represents a genuine risk factor rather than a statistical artifact. Our results extend the findings of Cooper et al. (2008) on asset growth and Fama and French (2015) on investment factors by demonstrating that operational efficiency metrics capture distinct aspects of firms' exposure to consumption-based systematic risk.

More broadly, our findings have important implications for understanding the economic mechanisms that generate cross-sectional return predictability within the rational pricing paradigm. The strong performance of overhead absorption efficiency across different market conditions and its resilience to risk-adjustment suggest that operational characteristics provide valuable information about firms' fundamental risk exposures that accounting variables alone cannot capture. By linking overhead absorption efficiency to consumption-based risk factors and the intertemporal marginal rate of substitution, we provide a unified framework for interpreting seemingly disparate return anomalies as manifestations of systematic consumption risk, advancing our understanding of how production-based and consumption-based models of asset pricing intersect in the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S. firms listed on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures.Overhead Absorption Efficiency signal relies on two key accounting variables. Invested capital (ICAPT) represents the total capital invested in a company's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This measure captures the total financial resources deployed in the business that require a return. Selling, general, and administrative expenses (XSGA) represent the operating expenses not directly tied to production, including costs related to sales activities, corporate management, administrative functions, marketing, and other overhead expenses necessary to support the firm's operations.construct the Overhead Absorption Efficiency signal by dividing invested capital (ICAPT) by selling, general, and administrative expenses (XSGA) for each firm-year observation. This ratio captures how efficiently a firm utilizes its overhead spending relative to its invested capital base—a higher ratio indicates that the firm supports more invested capital per dollar of overhead expense, suggesting greater operational efficiency in deploying capital without proportional increases in administrative costs. We calculate this measure annually using fiscal year-end values and require non-missing, positive values for both ICAPT and XSGA. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles within each year. Firms with missing or negative values for either component are excluded from the analysis for that year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OAE signal. Panel A plots the time-series of the mean, median, and interquartile range for OAE. On average, the cross-sectional mean (median) OAE is -7.90 (-3.23) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OAE data. The signal's interquartile range spans -9.15 to -1.30. Panel B of Figure 1 plots the time-series of the coverage of the OAE signal for the CRSP universe. On average, the OAE signal is available for 6.03% of CRSP names, which on average make up 6.61% of total market capitalization.

4 Does OAE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OAE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OAE portfolio and sells the low OAE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OAE strategy earns an average return of 0.41% per month with a t-statistic of 3.17. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.44% to 0.65% per month and have t-statistics exceeding 3.38 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.58, with a t-statistic of -10.59 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 499 stocks and an average market capitalization of at least $1,213 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 33 bps/month with a t-statistics of 2.55. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 21-51bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.54. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OAE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the OAE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OAE, as well as average returns and alphas for long/short trading OAE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the OAE strategy achieves an average return of 37 bps/month with a t-statistic of 2.52. Among these large cap stocks, the alphas for the OAE strategy relative to the five most common factor models range from 42 to 64 bps/month with t-statistics between 2.82 and 4.76.

5 How does OAE perform relative to the zoo?

Figure 2 puts the performance of OAE in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the OAE strategy falls in the distribution. The OAE strategy's gross (net) Sharpe ratio of 0.40 (0.39) is greater than 82% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OAE strategy (red line).[2] Ignoring trading costs, a \$1 invested in the OAE strategy

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$11.01 which ranks the OAE strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OAE strategy would have yielded \$9.88 which ranks the OAE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OAE relative to those. Panel A shows that the OAE strategy gross alphas fall between the 76 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OAE strategy has a positive net generalized alpha for five out of the five factor models. In these cases OAE ranks between the 90 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OAE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OAE with 210 filtered anomaly

of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OAE or at least to weaken the power OAE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OAE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OAE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OAE} OAE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OAE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OAE. Stocks are finally grouped into five OAE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OAE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OAE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OAE signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.21, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on OAE is 0.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the OAE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OAE strategy earns alphas that range from 26-54bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.72, which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OAE trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.41.

7 Does OAE add relative to the whole zoo?

Finally, we can ask how much adding OAE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OAE signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes OAE grows to $2659.68.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OAE is available.

8 Conclusion

This study provides compelling evidence that Overhead Absorption Efficiency (OAE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OAE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related accounting-based anomalies.

The key findings of our research underscore the economic importance of OAE as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.40 (0.39 net of transaction costs), indicating strong performance that is largely unaffected by implementation frictions. More notably, the strategy delivers monthly abnormal returns of 54 basis points (51 bps net) relative to the Fama-French five-factor model augmented with momentum, with highly significant t-statistics exceeding 4.4. These results suggest that OAE captures a distinct dimension of firm efficiency that is not fully explained by conventional risk factors.

Perhaps most importantly, OAE retains significant predictive power even when confronted with a comprehensive set of related accounting and market-based signals. The monthly alpha of 22 basis points (t-statistic = 2.41) relative to twelve factors—including the six standard factors and six closely related strategies such as gross profitability, operating leverage, and cash productivity—demonstrates that OAE contains unique information about future returns beyond what is captured by existing anomalies in the factor zoo.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, OAE represents a meaningful addition to our understanding of how operational efficiency metrics relate to expected returns, potentially offering insights into the economic mechanisms linking firm fundamentals to market valuations. For practitioners, the strong net-of-cost performance suggests that OAE-based strategies may offer implementable alpha opportunities, particularly given the signal's resilience to transaction costs.

Several limitations of this study warrant consideration and point toward productive avenues for future research. First, while our analysis demonstrates the statistical and economic significance of OAE, the underlying economic mechanism driving the return predictability remains an open question. Future research should investigate whether the OAE premium reflects compensation for systematic risk, market frictions, or behavioral biases. Second, the stability and persistence of the OAE effect across different market regimes and international markets deserves further examination. Third, understanding how OAE interacts with firm characteristics such as size, industry classification, and business cycle sensitivity could provide deeper insights into when and where the signal is most effective.

Future research directions should also explore the optimal combination of OAE with other efficiency metrics to construct more powerful composite signals. Additionally, investigating the real-time implementability of OAE strategies, including the impact of data availability lags and portfolio rebalancing frequencies, would provide valuable insights for practical application. Finally, examining whether firms can strategically manipulate their overhead absorption metrics and how such behavior might affect the signal's predictive power represents an important area for further investigation.

In conclusion, this paper establishes Overhead Absorption Efficiency as a robust and economically meaningful predictor of stock returns that survives stringent empirical tests and offers both theoretical insights and practical value to the asset pricing literature.

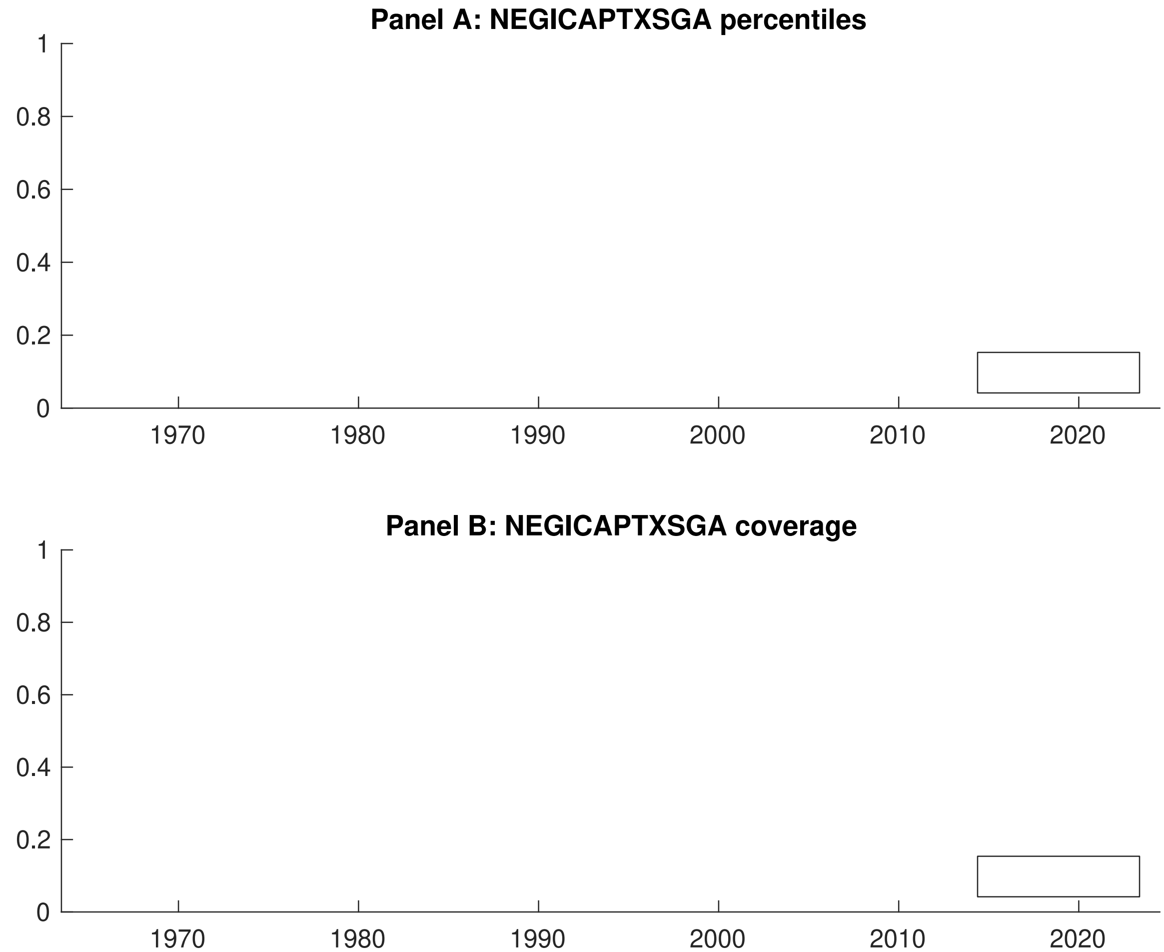


Figure 1: Times series of OAE percentiles and coverage.
This figure plots descriptive statistics for OAE. Panel A shows cross-sectional percentiles of OAE over the sample. Panel B plots the monthly coverage of OAE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OAE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OAE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.35	0.54	0.71	0.63	0.75	0.41
	[1.84]	[3.08]	[3.80]	[3.57]	[4.38]	[3.17]
α_{CAPM}	-0.26	-0.05	0.08	0.04	0.18	0.44
	[-3.09]	[-0.90]	[1.32]	[0.63]	[2.89]	[3.38]
α_{FF3}	-0.38	-0.15	0.14	0.12	0.27	0.65
	[-5.11]	[-3.05]	[2.54]	[2.20]	[4.56]	[5.64]
α_{FF4}	-0.37	-0.11	0.19	0.12	0.25	0.62
	[-4.90]	[-2.31]	[3.41]	[2.12]	[4.14]	[5.29]
α_{FF5}	-0.37	-0.16	0.18	0.11	0.19	0.56
	[-4.90]	[-3.26]	[3.06]	[1.91]	[3.24]	[4.84]
α_{FF6}	-0.37	-0.13	0.22	0.10	0.18	0.54
	[-4.74]	[-2.62]	[3.76]	[1.84]	[3.01]	[4.62]
Panel B: Fama and French (2018) 6-factor model loadings for OAE-sorted portfolios						
β_{MKT}	1.08	1.03	1.02	0.98	0.96	-0.12
	[58.00]	[86.18]	[73.29]	[72.05]	[68.39]	[-4.20]
β_{SMB}	-0.03	0.00	0.06	0.03	0.04	0.07
	[-1.07]	[0.14]	[2.84]	[1.73]	[2.03]	[1.71]
β_{HML}	0.33	0.25	-0.20	-0.19	-0.25	-0.58
	[9.14]	[10.82]	[-7.33]	[-7.33]	[-9.19]	[-10.59]
β_{RMW}	-0.05	0.05	-0.05	0.11	0.21	0.26
	[-1.37]	[1.97]	[-1.81]	[4.04]	[7.72]	[4.75]
β_{CMA}	0.05	0.02	-0.04	-0.11	0.03	-0.02
	[0.99]	[0.45]	[-1.07]	[-2.84]	[0.76]	[-0.27]
β_{UMD}	-0.01	-0.04	-0.06	0.00	0.02	0.03
	[-0.51]	[-3.68]	[-4.26]	[0.21]	[1.14]	[0.91]
Panel C: Average number of firms (n) and market capitalization (me)						
n	499	585	603	659	873	
me ($\$10^6$)	1213	1630	1918	2206	1609	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the OAE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.41	0.44	0.65	0.62	0.56	0.54
			[3.17]	[3.38]	[5.64]	[5.29]	[4.84]	[4.62]
Quintile	NYSE	EW	0.34	0.26	0.41	0.49	0.57	0.64
			[2.48]	[1.91]	[3.49]	[4.16]	[4.96]	[5.46]
Quintile	Name	VW	0.42	0.43	0.64	0.59	0.56	0.53
			[3.16]	[3.27]	[5.49]	[4.99]	[4.74]	[4.39]
Quintile	Cap	VW	0.33	0.36	0.57	0.52	0.47	0.44
			[2.55]	[2.77]	[4.92]	[4.44]	[4.04]	[3.71]
Decile	NYSE	VW	0.53	0.55	0.76	0.76	0.73	0.72
			[3.17]	[3.24]	[4.88]	[4.76]	[4.54]	[4.46]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.39	0.42	0.60	0.59	0.52	0.51
			[3.06]	[3.21]	[5.25]	[5.10]	[4.54]	[4.45]
Quintile	NYSE	EW	0.21	0.13	0.26	0.31	0.37	0.41
			[1.54]	[0.96]	[2.17]	[2.61]	[3.19]	[3.54]
Quintile	Name	VW	0.40	0.41	0.59	0.57	0.52	0.50
			[3.05]	[3.05]	[5.09]	[4.86]	[4.40]	[4.24]
Quintile	Cap	VW	0.32	0.35	0.53	0.51	0.44	0.42
			[2.44]	[2.66]	[4.60]	[4.37]	[3.80]	[3.65]
Decile	NYSE	VW	0.51	0.50	0.70	0.70	0.65	0.65
			[3.05]	[2.98]	[4.45]	[4.43]	[4.09]	[4.09]

Table 3: Conditional sort on size and OAE

This table presents results for conditional double sorts on size and OAE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OAE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OAE and short stocks with low OAE .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	OAE Quintiles (L)	(2)	(3)	(4)	(H)	OAE Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.51	0.71	0.83	0.88	0.76	0.24	0.08	0.24	0.27	0.49	0.50
	[2.37]	[3.08]	[3.27]	[3.36]	[2.57]	[1.34]	[0.44]	[1.54]	[1.65]	[3.20]	[3.19]
(2)	0.64	0.76	0.74	0.79	0.88	0.24	0.18	0.35	0.38	0.42	0.44
	[2.78]	[3.41]	[3.18]	[3.28]	[3.56]	[1.72]	[1.27]	[2.81]	[3.01]	[3.33]	[3.48]
(3)	0.68	0.69	0.73	0.84	0.83	0.16	0.14	0.31	0.36	0.34	0.38
	[2.99]	[3.20]	[3.36]	[3.72]	[3.72]	[1.16]	[1.05]	[2.44]	[2.78]	[2.66]	[2.92]
(4)	0.52	0.59	0.73	0.77	0.91	0.39	0.41	0.58	0.60	0.61	0.62
	[2.39]	[2.86]	[3.49]	[3.68]	[4.42]	[2.81]	[2.96]	[4.52]	[4.55]	[4.58]	[4.61]
(5)	0.33	0.53	0.69	0.59	0.71	0.37	0.42	0.64	0.62	0.55	0.53
	[1.76]	[3.05]	[3.69]	[3.47]	[4.20]	[2.52]	[2.82]	[4.76]	[4.46]	[4.03]	[3.83]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	OAE Quintiles Average n (L)	(2)	(3)	(4)	(H)	OAE Quintiles Average market capitalization ($\$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	375	378	377	375	362	35	34	33	29	22
(2)	100	101	101	101	100	52	52	52	51	52
(3)	69	69	69	69	69	86	87	87	87	87
(4)	53	54	54	54	54	174	174	176	179	179
(5)	47	47	48	47	48	1009	1326	1715	1477	1319

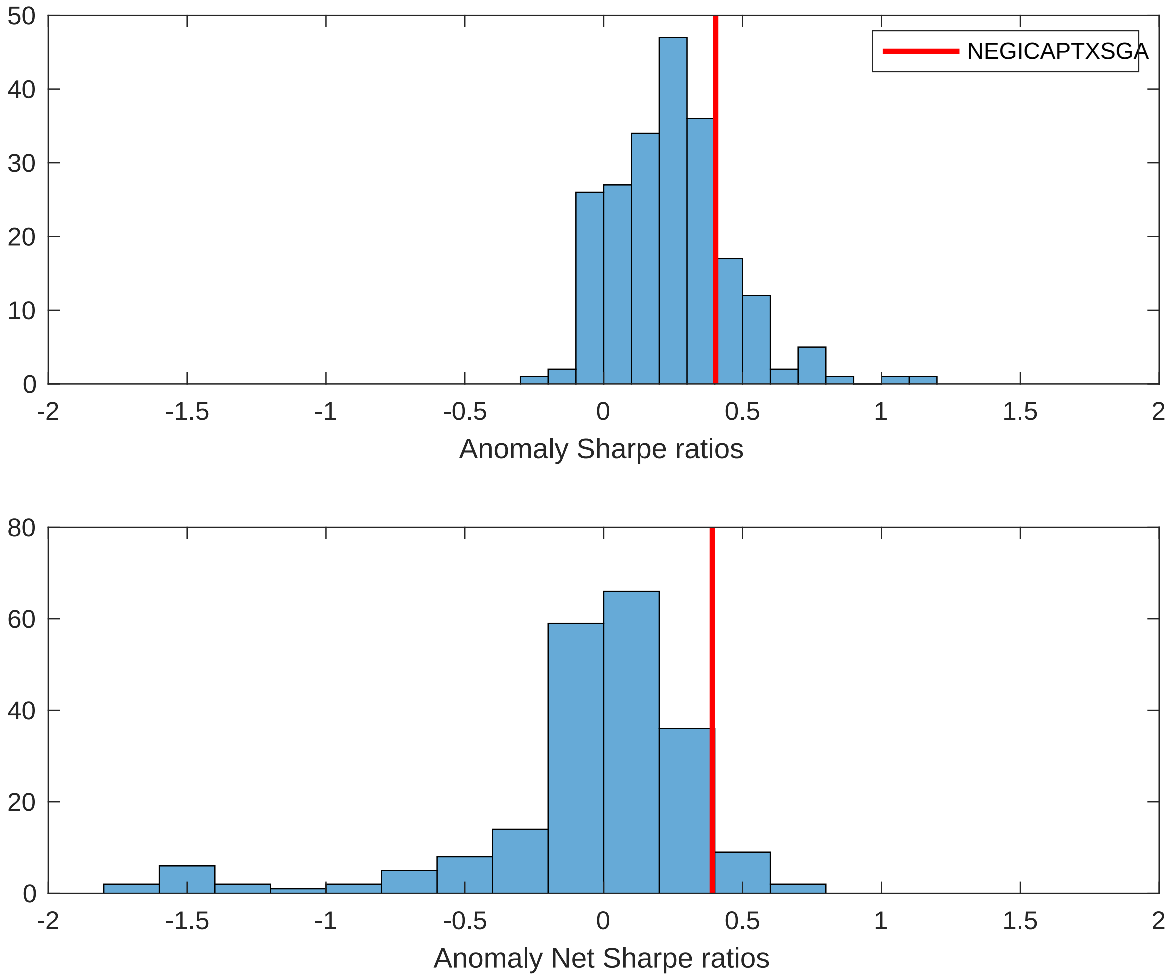


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OAE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

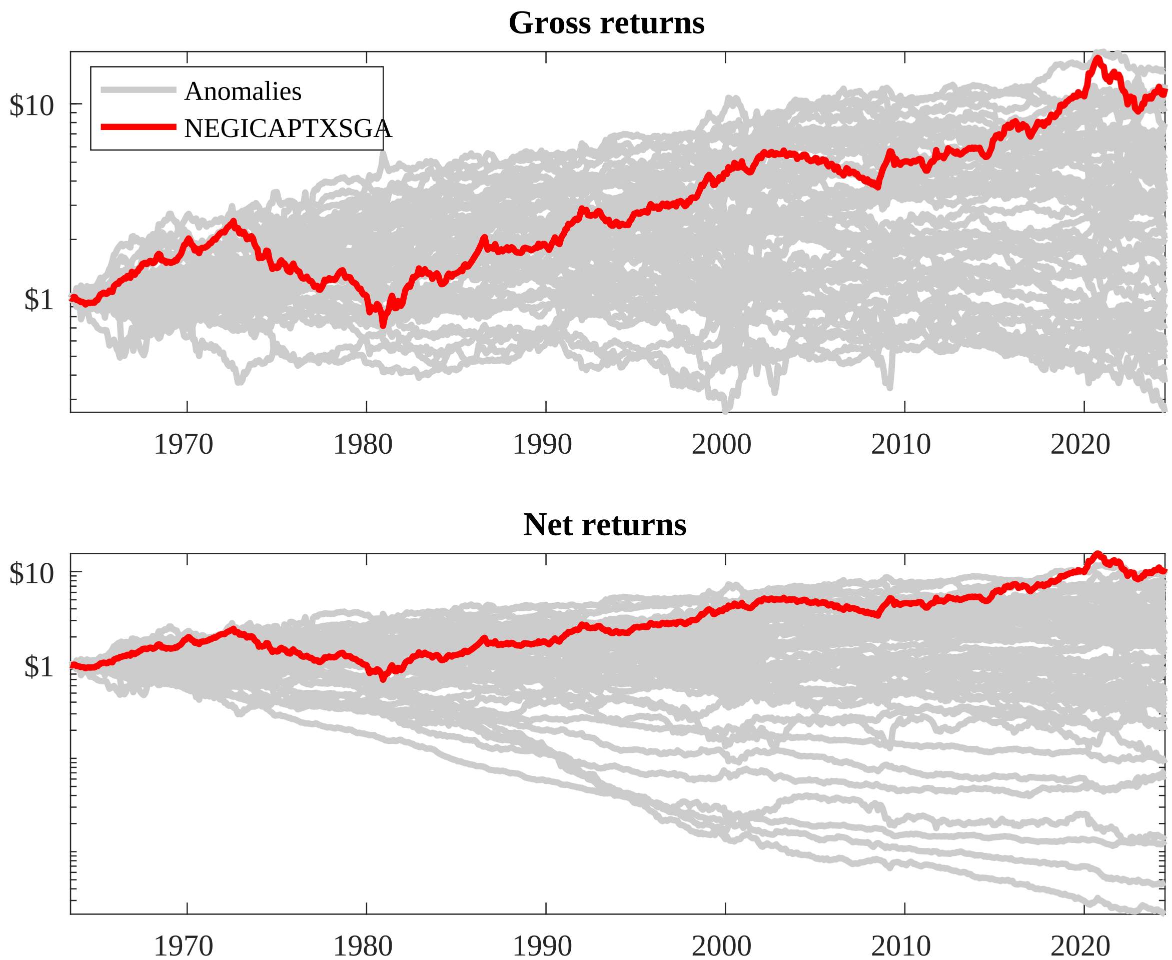
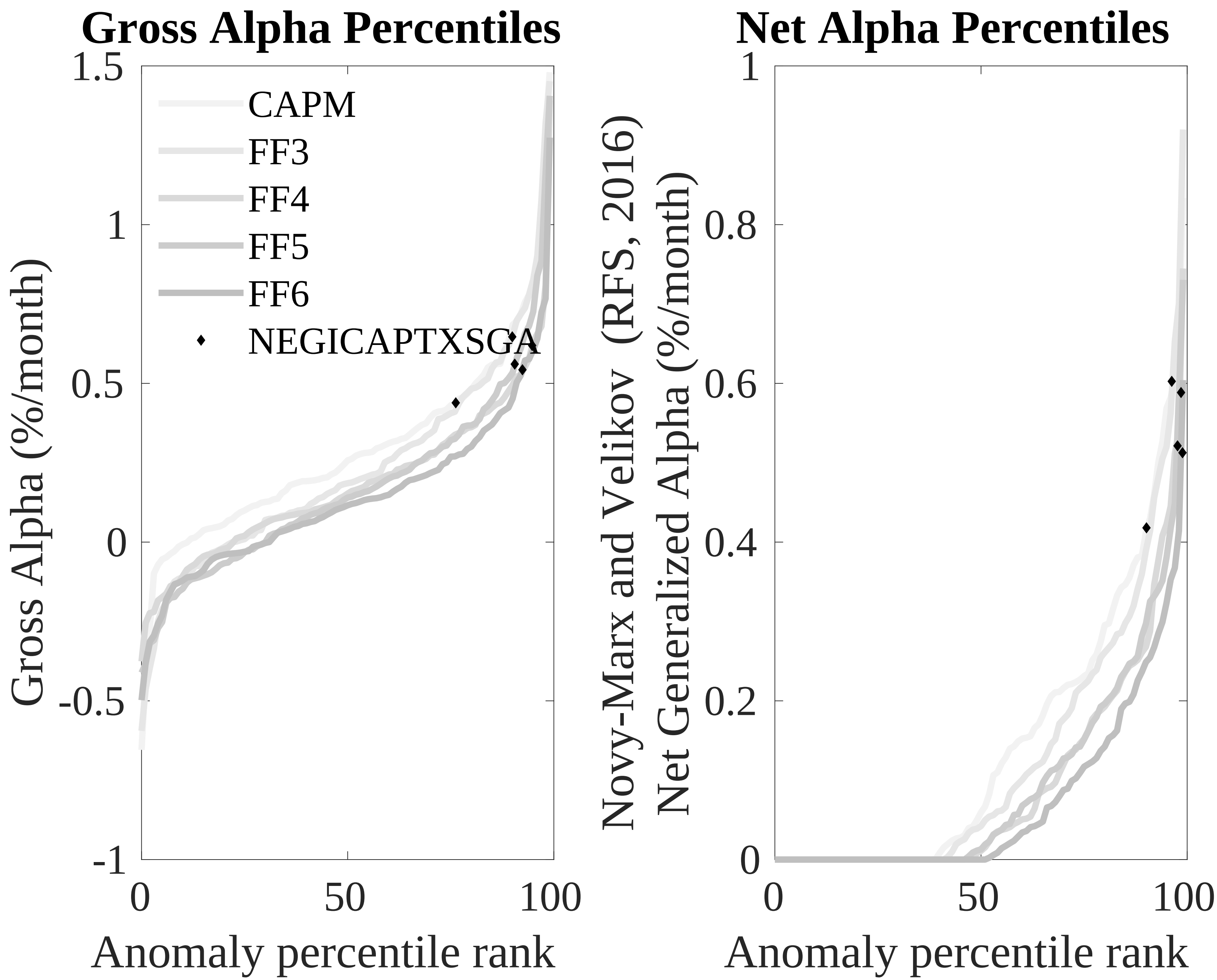


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OAE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGICAPTXSGA
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

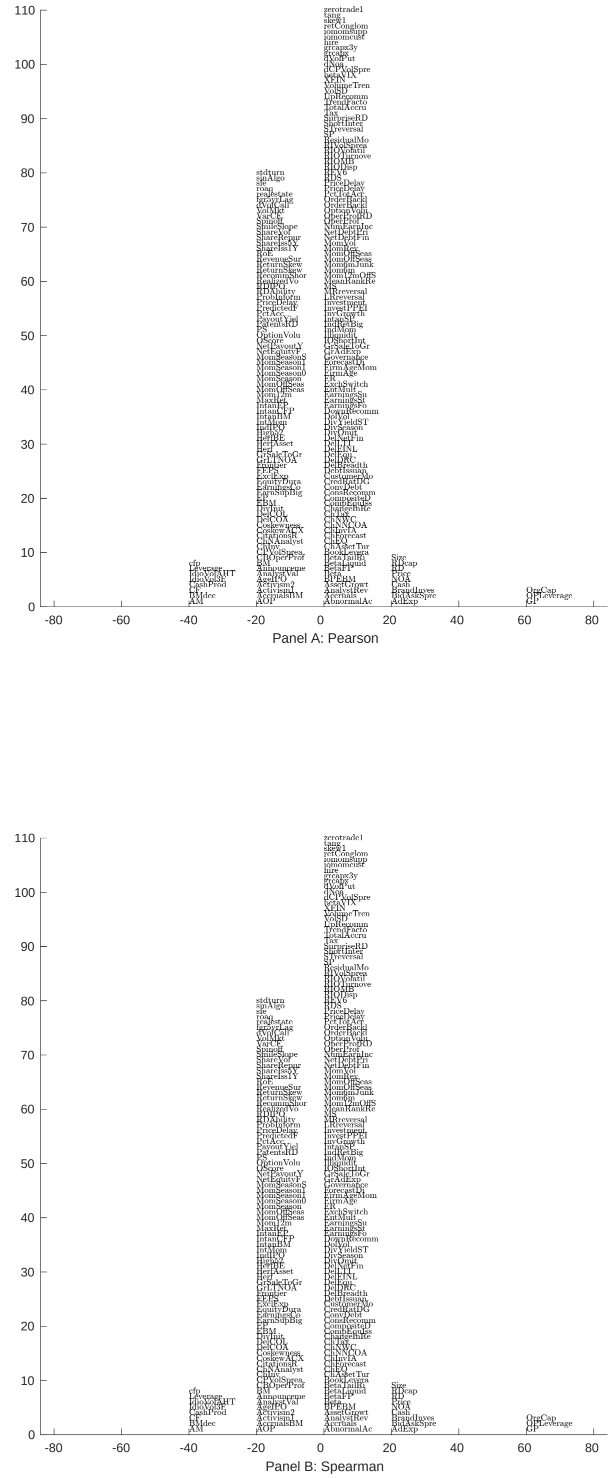


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with OAE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

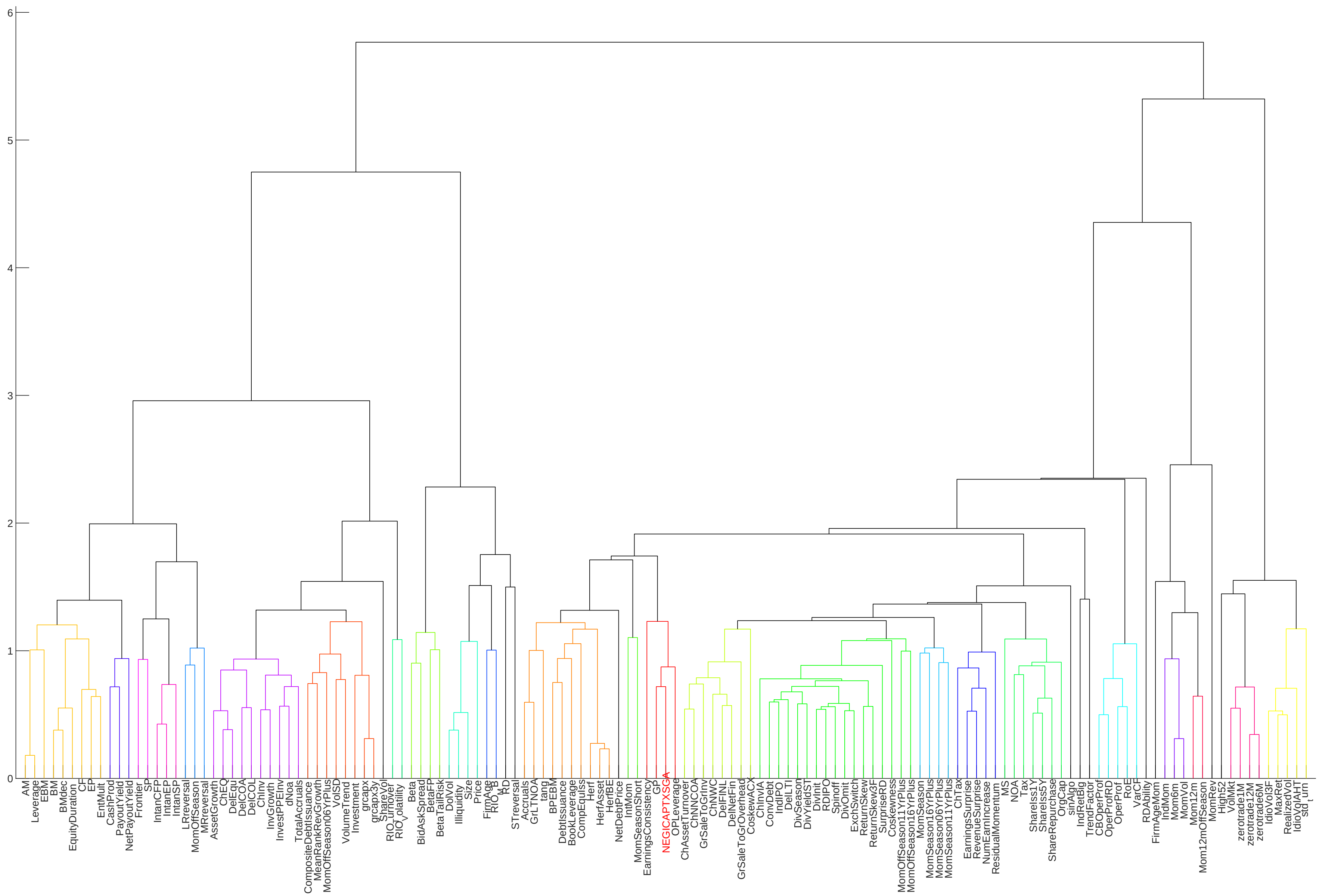


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

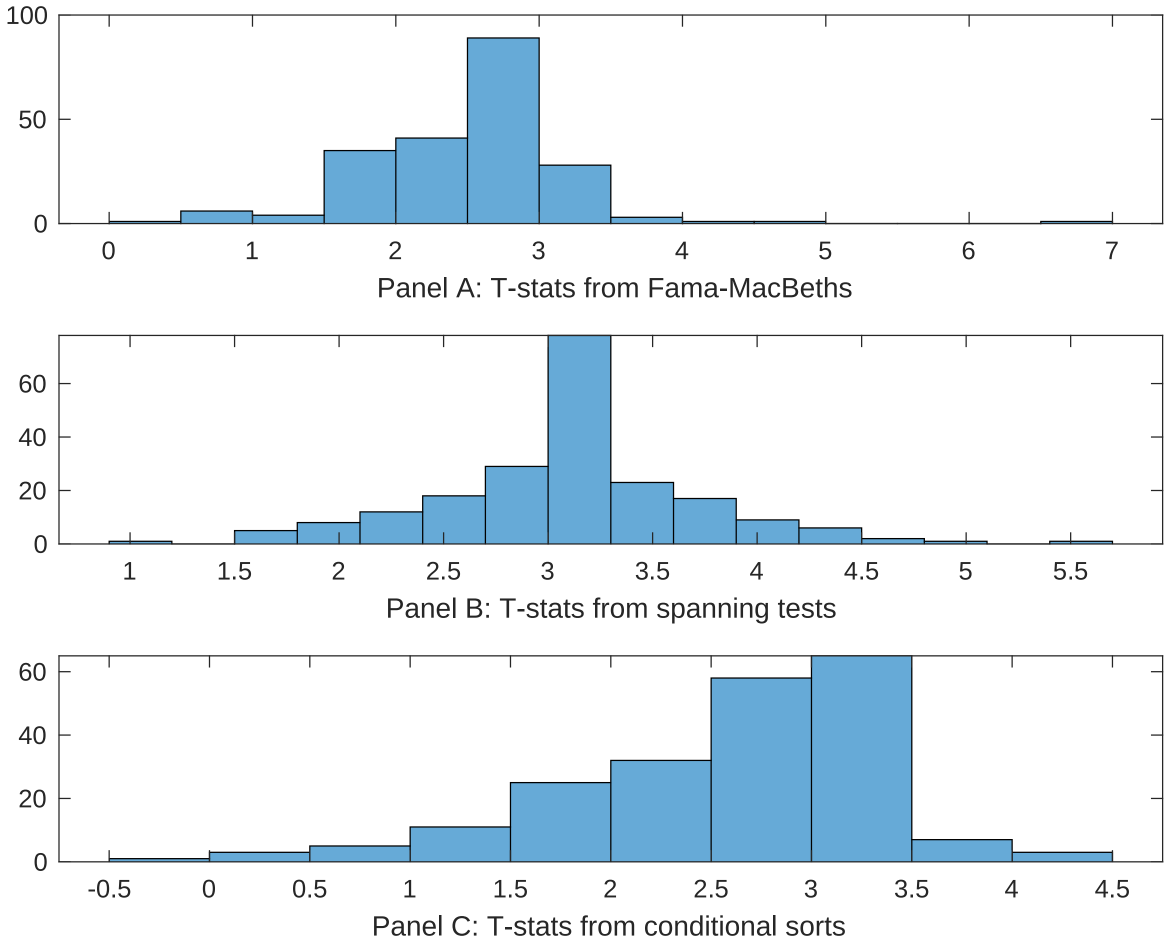


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of OAE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OAE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OAE} OAE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OAE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OAE. Stocks are finally grouped into five OAE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OAE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OAE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OAE} OAE_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.81	0.11	0.11	0.97	0.12	0.12	0.48
	[2.85]	[4.70]	[4.82]	[4.12]	[5.21]	[5.01]	[1.77]
OAE	0.12	0.74	0.14	0.16	0.13	0.13	0.45
	[0.21]	[1.69]	[2.94]	[3.43]	[2.68]	[2.74]	[0.88]
Anomaly 1	0.94						0.11
	[5.93]						[7.45]
Anomaly 2		0.11					-0.48
		[2.96]					[-1.34]
Anomaly 3			0.48				0.21
			[3.26]				[2.42]
Anomaly 4				0.36			0.18
				[5.64]			[1.58]
Anomaly 5					0.20		-0.45
					[1.97]		[-0.58]
Anomaly 6						0.50	-0.19
						[2.88]	[-1.99]
# months	732	732	727	727	727	727	727
$\bar{R}^2(\%)$	1	0	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the OAE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OAE} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are gross profits / total assets, Operating leverage, Total assets to market, Book to market using December ME, Cash Productivity, Market leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.26	0.47	0.54	0.47	0.54	0.53	0.22
	[2.72]	[4.19]	[4.54]	[4.15]	[4.55]	[4.48]	[2.41]
Anomaly 1	76.99						79.59
	[19.54]						[15.94]
Anomaly 2		41.71					20.40
		[9.26]					[4.55]
Anomaly 3			-2.16				-13.24
			[-0.43]				[-0.90]
Anomaly 4				-54.43			-25.49
				[-7.48]			[-3.99]
Anomaly 5					-13.77		37.69
					[-2.35]		[6.93]
Anomaly 6						-6.48	22.97
						[-1.33]	[1.60]
mkt	-13.08	-11.48	-11.74	-11.25	-13.42	-11.14	-10.15
	[-5.71]	[-4.29]	[-4.03]	[-4.12]	[-4.66]	[-3.83]	[-4.29]
smb	2.58	-11.20	8.10	26.07	8.66	8.57	-1.47
	[0.78]	[-2.56]	[1.95]	[5.62]	[2.11]	[2.08]	[-0.35]
hml	-24.00	-35.03	-53.88	-0.49	-46.21	-49.05	-24.31
	[-5.06]	[-6.15]	[-6.74]	[-0.05]	[-6.64]	[-6.33]	[-2.88]
rmw	-17.94	3.45	26.55	14.33	25.55	26.53	-33.67
	[-3.58]	[0.60]	[4.78]	[2.57]	[4.63]	[4.81]	[-6.27]
cma	10.05	-10.28	-3.23	0.41	-1.24	-3.82	4.79
	[1.54]	[-1.34]	[-0.40]	[0.05]	[-0.15]	[-0.47]	[0.75]
umd	0.36	1.69	1.87	4.12	-0.59	0.78	11.36
	[0.16]	[0.63]	[0.60]	[1.52]	[-0.19]	[0.25]	[4.31]
# months	732	732	728	728	728	728	728
$\bar{R}^2$(%)	51	33	25	31	26	25	56

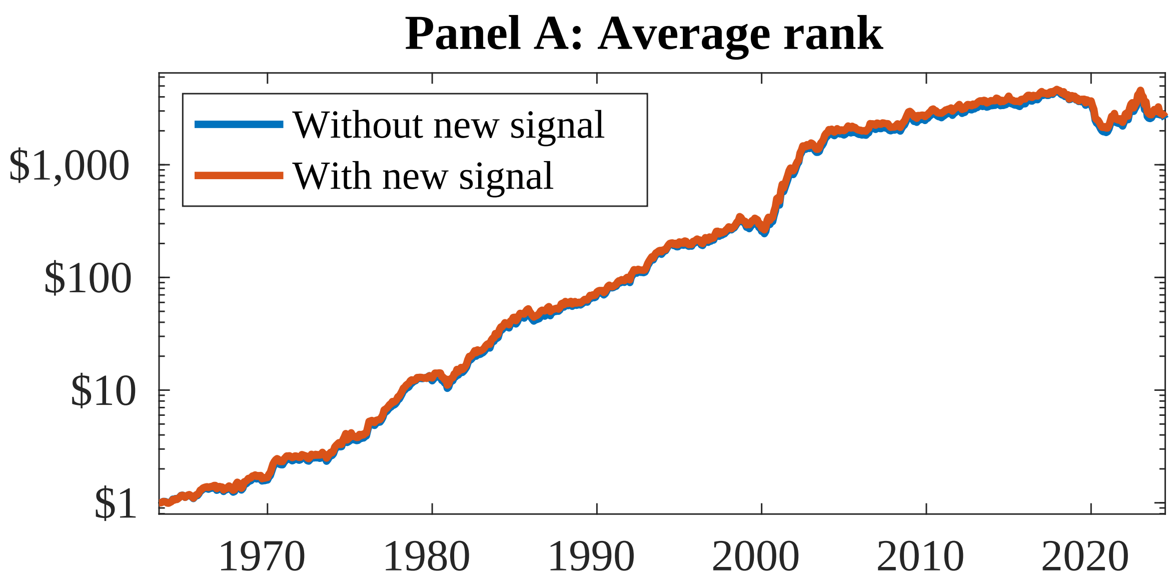


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OAE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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